

Building an Aurora Visibility Forecast Tool Using Machine Learning Models

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1. Introduction

Auroras are breathtaking natural light displays created when energetic particles from the solar wind collide with oxygen and nitrogen molecules in Earth's upper atmosphere, causing them to glow in vibrant shades of green, purple, and red. These atmospheric collisions naturally form a ring-shaped band called the auroral oval, centered around Earth's magnetic poles. Under typical conditions, this ring sits high in the Arctic Circle, making northern regions like Scandinavia, Canada, Alaska, and Iceland the most common places to spot the lights.

To quantify solar activity and track geomagnetic disturbances, space scientists use the Kp index, which is a global geomagnetic activity scale ranging from 0 to 9. Low Kp values (0 to 2) indicate quiet space weather, keeping the aurora faint and locked close to the polar caps. Higher values (Kp 5 and above) represent geomagnetic storms, which cause the auroral oval to expand southward toward lower latitudes where millions more people can see it.

Since geomagnetic activity can change rapidly, it can be difficult to predict the occurrence and strength of geomagnetic storms and determine if auroras are visible in certain locations. Our project addresses these primary questions: (1) can we predict the occurrence and/or strength of a geomagnetic storm? and (2) can we predict the probability of seeing an aurora from a certain location? To help answer these questions, we developed machine learning models to predict the Kp index at three time horizons: 0-hour nowcast, 3-hour forecast, and 6-hour forecast.

Our motivation for this project stems from our shared interest in space weather and auroras, and how we were all able to witness one by chance. Stargazers and others interested in auroras have a difficult time viewing them outside of the typical high-latitude regions, often resorting to monitoring space weather conditions to determine if an aurora will occur. Beyond aurora viewing, predicting space weather has a broader real-world importance because strong geomagnetic storms can disrupt space-based technologies such as satellite communications and GPS navigation, and can contribute to fluctuations in electrical power grids. Because of this, our project aims to make this information more accessible and easier to interpret for these users, both for aurora viewing and for a deeper understanding of disruptive space-weather events.

Our solution also combines geomagnetic activity prediction with a visualization of the auroral oval, which is an oval-shaped region centered around the magnetic poles that indicates where auroras have the highest chance of occurring. We use the predicted Kp index from our model and translate it into generating the auroral oval, with it expanding towards lower latitudes as Kp increases. Our visualization also accounts for the asymmetric nature of the oval based on the current time. Through our Streamlit application, users can select a location to examine the auroral probability, the visualization of the auroral oval, the predicted Kp index, and the forecast time. Additionally, we provide a way for users to input a custom Kp index and time to generate what the auroral oval would look like under those conditions. Unlike existing applications such as "My Aurora Forecast," which uses already-existing forecasts and space-weather information, our app generates its own Kp prediction using machine learning to make a prediction and dynamically generate the corresponding auroral oval. This allows users to explore not only the current forecast, but also how changes in geomagnetic activity and time can affect the size and shape of the oval.

2. Related Work

Machine learning methods have been used by several researchers to predict geomagnetic storm activity. Zhelavskaya et al. (2019) employed linear regression, Extreme Gradient Boosting, and a feedforward

neural network (FNN) to predict Kp index as a continuous value at prediction horizons ranging from 0-hour (“nowcast”) to 12-hour. They primarily used time-lagged features derived from interplanetary magnetic field (IMF) measurements, solar wind speed, and proton density, looking back up to 9 hours from the current time. Ultimately, they concluded that FNN out-performed the other methods, noting that the error rate increased for all models as the forecast period lengthened. Conde et al. (2023) utilized a long short-term memory (LSTM) neural network to predict values for a geomagnetic activity index called the symmetric disturbance in the horizontal component of Earth’s magnetic field (SYM-H). The SYM-H index is similar to the Kp index, with the main difference being that SYM-H is gathered with 1-minute resolution rather than every 3 hours. Conde and his colleagues split up their dataset into individual storms occurring between 1998 and 2018 and distributed them into training, test, and validation sets. They used a smaller set of variables than Zhelavskaya et al., electing to use components of the IMF and arriving at a model that used 75 look-backs in 5-minute increments. Khaled et al. (2023) conducted a survey of nine other works that used various neural network models (including LSTM) to predict Kp values. Some of these works predicted Kp as a continuous value, while others attempted to classify storms by their strength.

Similar to Zhelavskaya et al. (2019) and Conde et al. (2023), our work seeks to predict the Kp index as a continuous value. One of the objectives of our project is to construct a lightweight model that can be deployed and quickly adjusted for real-time predictions. Thus, we closely followed Zhelavskaya et al.’s methods for the linear regression model, taking a similar approach in terms of creating time-lagged features from the solar activity variables. We saw in Conde et al.’s work that LSTM networks could be effective when predicting geomagnetic disturbances, but we chose to use it as a regressor, rather than a classifier. Our LSTM model depends on sequenced look-backs like Conde, but we used hourly blocks instead of 5-minute increments.

3. Data and Methodology

3.1 Data Sources

In our development process, we utilized both historical and real-time data relating to geomagnetic storm observations. Space weather prediction relies on continuous monitoring of interplanetary magnetic fields, solar wind speed, proton density, and geomagnetic indices captured by spacecraft observations. The historical variables utilized in this project were extracted from the National Aeronautics and Space Administration (NASA) OMNIWeb dataset and cover several fundamental domains of space plasma physics. The IMF measurements include overall magnetic magnitude, vector-averaged directional components (Bx, By, and Bz in Geocentric Solar Magnetospheric coordinates), and their corresponding standard deviations to track vector fluctuations. Additionally, these measurements have been available both on the 1-hour and 1-minute resolution since the 1960s and 1980s, respectively. Taking inspiration from Zhelavskaya et al. (2019), we gathered 1-minute resolution data for about 20 years, from 2000 to mid-2026. After collection, the size of the dataset we used for training contains around 14 million rows and 17 columns.

Another historical dataset we used comes from Aurorasaurus, a website that contains user-submitted observations of auroras at specific locations and times. The dataset we used contains observations from 2014 to 2025, containing around 22,000 rows and 33 columns. Although this data wasn’t directly used in our model, it did aid in helping our team understand and visualize the auroral oval, and why certain locations have a higher concentration of positive aurora observations than others. As for our real-time dataset, we continuously pull data from the National Oceanic and Atmospheric Administration (NOAA) Space Weather Prediction Center. They provide endpoints that return JSON data containing the same satellite measurement features we use from OMNIWeb. The difference between the two is that OMNIWeb data can lag behind the current time by a couple weeks, while the NOAA data is updated every minute, and that it only lags behind at most 10 minutes behind the current time. This allows our trained model to be applied to current space weather conditions to essentially yield real-time predictions and aurora forecasts.

3.2 Data Cleaning and Exploration

Of course, we had to take some careful steps with preparing our data before we conducted time-series training and testing for our model, the most crucial data being the historical OMNIWeb dataset. Exploratory data analysis was conducted to examine the distribution of each solar wind and magnetic field feature, as well as examine any outliers and any null values that occur within the dataset. A few variables were found to be right-skewed in distribution, which we've taken into account during feature engineering. As for null values, we found that approximately 20% of rows contained not-a-number (NaN) values, with the longest continuous run of null values going as long as 3 weeks. Considering this and the time-window of the whole dataset (~26 years), we decided to linearly interpolate any continuous NaN runs less than an hour, then drop any remaining null rows. This process was performed on each set of data (training, validation, testing) independently to avoid any data leakage.

3.3 Training and Validation Setup

To avoid data leakage in the time series, we split our training, validation, and test sets chronologically before any imputation or feature engineering occurred. We chose the most recent 3 years for our test set: June 30, 2023 to June 30, 2026. The test set contained 1.5 million records before feature engineering, representing approximately 11% of the data. The validation set was made up of 1.8 million records in the range January 1, 2020 to June 30, 2023, approximately 15% of the data. The training set covered the remaining time period of our data and contained ~10 million records. This was done to ensure that no future data would be used to predict past data. We performed our parameter searches against our held-out validation set rather than splitting into further folds with GridSearchCV. We accepted the trade-off of having a potentially optimistic validation score in the interest of having realistic results against the test set and new data.

3.4 Feature Engineering

In order to ensure our features could work with linear models, we applied log transformations to the variables that displayed right-skewed distributions. The magnitude of the interplanetary magnetic field, flow speed, and proton density had skewness values ranging from 0.09 to 3.30. As we were working with time series data, we wanted to provide meaningful model inputs that would capture historical patterns. Similar to Zhelavskaya et al. (2019), we created time-lagged features based on the solar wind and magnetic field variables. Taking the minute-resolution data for each variable, we calculated the average, minimum, and maximum over 4 hourly look-backs for each target value. Our target variable, the Kp index, is only reported by GFZ Helmholtz Centre for Geosciences (n.d.) every 3 hours, so we aggregated the time-lagged features into 3-hour increments starting at 00:00 each day. Together with time-based features to account for seasonal variations, our feature engineering yielded 76 total features. Further, we established a threshold that dropped any 3-hour intervals that were missing more than 5 minutes, including in any of the 4 look-backs. After creating the aggregated features and dropping the intervals over the threshold, we were left with 55,234 rows in the training set, 9,713 rows in the validation set, and 8,158 rows in the test set. See **Appendix A** for the list of features used to train the Lasso regression model and their coefficient values.

To prepare our data for LSTM neural networks, we created 24-hour look-back sequences from each target Kp value. Each sequence included the average, minimum, and maximum of the 6 solar variables, calculated from the minute-level data. The resulting feature array for the training set contained 50,497 sequences. The validation and test sets consisted of 9,041 sequences and 7,558 sequences, respectively.

3.5 Machine Learning Methods

In our project, we seek to predict the strength of geomagnetic storms to provide users with forecasts about aurora visibility. We chose the Kp index as our target variable since it is widely used in the machine learning literature and by NOAA (n.d.) in their own aurora visibility projections. Since the Kp index has a scale of 0 to 9 including fractional values, we approached Kp prediction as a regression task with a continuous target variable. The Kp index is only reported every 3 hours, so the value is held constant for

the intermediate hours. Within the scope of our project, we decided to build models for three time horizons that align to the Kp intervals: a 0-hour nowcast model, a 3-hour forecast model, and a 6-hour forecast model.

In developing a machine learning model, we had two goals: (1) build a model with an acceptable level of prediction error for aurora visibility, and (2) select an easy-to-maintain model that would execute smoothly in a real-time data pipeline. We explored three different model types: linear regression, Lasso regression, and LSTM neural networks. We chose linear regression because Zhelavskaya et al. (2019) demonstrated promising results and we thought it would be a good candidate for a real-time data stream. Given the number of features after engineering (76), our next inclination was to include Lasso because of its ability to identify important features and shrink the coefficients for less important ones. Finally, we decided to explore LSTM based on its prevalence in the space weather modeling literature, such as Conde et al. (2023) and Khaled et al. (2023), and its ability to capture non-linear relationships, which are known to occur in geomagnetic storm data.

3.6 Hyperparameter Tuning

We trained and tuned all three models for the 0-hour nowcast, using the Kp index of the current 3-hour interval as the target variable. For Lasso regression, we performed a grid search on the alpha parameter, finding the optimal value at 0.0008. For LSTM, we explored values for three parameters: number of layers, the hidden size, and the learning rate. The range of values we searched is shown in **Table 1**. We found that 1 layer with 64 hidden units and a learning rate of 10^{-3} yielded the best performance. As Ordinary Least Squares (OLS) linear regression does not have parameters that control regularization or complexity, we did not need to run a grid search for it.

Hyperparameter	Search Grid	Selected Value
Number of layers	[1, 2, 3, 4]	1
Hidden size	[16, 64, 256]	64
Learning rate	[10^{-4} , 10^{-3} , 10^{-2}]	10^{-3}

Table 1 - Ranges of Hyperparameter Values Searched for LSTM (0-Hour Nowcast)

3.7 Methodology for Translating Predicted Values into Aurora Visibility

To convert Kp index model outputs into visual geographic predictions, the model predictions are translated into an auroral visibility spatial boundary. Rather than relying on a static circular expansion around the geographic poles, the visual model utilizes a dynamic oval representation influenced by geomagnetic activity modeling centered around the geomagnetic poles (Feldstein, 1963; Holzworth & Meng, 1975).

The translation pipeline takes the predicted Kp value alongside temporal variables such as time of day, date, and subsolar longitude. Spherical trigonometry adjusts for Earth's magnetic pole tilt by mapping geographic coordinates into magnetic latitude (Hapgood, 1992). A periodic cosine function tied to solar longitude handles day-night asymmetry, shifting the oval five degrees equatorward on the nightside while thinning the dayside arc (Holzworth & Meng, 1975). As the predicted Kp increases, the auroral oval expands southward in latitude and broadens in spatial extent (Feldstein, 1963; Newell et al., 2014). To compute the location-based probability, the model calculates the given coordinate's distance from the peak latitudes using a Gaussian distribution (Machol et al., 2012; Newell et al., 2014). The calculated oval geometries reflect geomagnetic local time shifts, adjusting the dynamic band where visual auroral precipitation is most likely to occur over geographic map coordinates (Machol et al., 2012; Newell et al., 2014).

4. Model Results

We found that LSTM performed the best of our machine learning models for the 0-hour nowcast prediction task, with a Root Mean Squared Error (RMSE) of 0.69 and a Mean Absolute Error (MAE) of 0.53 on the held-out test set. The Lasso regression model performed the second best, with an RMSE of

0.76 and an MAE of 0.58 on the test set. OLS regression achieved virtually the same score as Lasso. The specific scores for each model are displayed in **Table 2**. Given our goal to deploy a light-weight model for real-time predictions and the comparable MAE scores between LSTM and Lasso, we chose the Lasso regression model for further development into 3- and 6-hour forecast models. An MAE of 0.58 means that our Lasso regression model differed from the observed Kp values by approximately 0.58 Kp units on average, which is 6.4% of the Kp scale range (0 to 9). Though not a direct comparison, for reference, Zhelavskaya et al. (2019) reported an MAE of 0.38 for their FNN nowcast model. While there was some overlap, it should also be noted that their models were trained on data from a different time period: 1998–2017.

According to the NOAA (n.d.), Kp values 0–5 have a low probability of producing a visible aurora. When the Kp index is greater than 5, it takes 1 whole Kp unit to meaningfully shift visibility between states. For example, a Kp index of 5 can produce a visible aurora from northern Michigan. If the Kp index goes up to 6, it can be visible in New York. Given that our solution is meant for use by casual hobbyists and does not have a direct impact on commercial operations, we decided that the MAE for the Lasso model was acceptable. In terms of processing time, our pipeline with real-time inferences from the Lasso regression model ran in 0.6 seconds end-to-end (local machine). **Figure 1** provides a good visualization of the performance of the Lasso model, showing the predictions closely following the actual Kp values over a 3-week period in 2025, which was part of the test set. We chose 3 weeks so that the steps and separation in the plot would be more discernable to the reader.

Rank	Model	RMSE (Test Set)	MAE (Test Set)	R ² (Test Set)
1	LSTM	0.6911	0.5256	0.7424
2	Lasso Regression	0.7559	0.5751	0.6994
3	OLS Regression	0.7566	0.5752	0.6989

Table 2 - Holdout Performance by RMSE (0-Hour Nowcast)

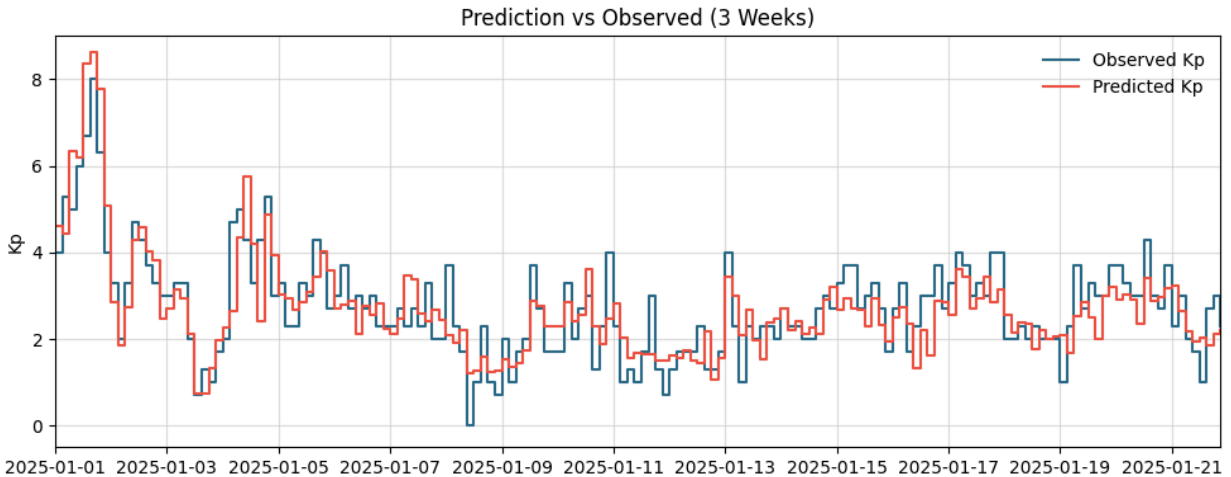


Figure 1 - Prediction vs. Observed Step Plot for the Lasso Regression Model (0-Hour Nowcast)

To develop the forecast models, we used the same features, but shifted the target Kp value forward to a future interval. For example, for a row containing the average flow speed for the Kp interval timestamped at 03:00 on a given day, the target of the 3-hour forecast model was the Kp value at the row timestamped 06:00. Consistent with Zhelavskaya et al.'s (2019) experience, the performance of our model decreased as we increased our prediction horizon. **Figure 2** shows the RMSE for the Lasso models for each horizon. The RMSE for the 3-hour model increased by about 0.20 relative to the nowcast baseline, whereas the 6-hour model increased by about 0.30. The graph also shows relative performance versus a Kp persistence model, which simply takes the last known Kp value as the prediction. We were initially

surprised to see the 6-hour model beat the persistence model while the 3-hour model did not. This suggests that the persistence model is difficult to beat for short horizons, but for forecasts farther into the future, our model has superior performance.

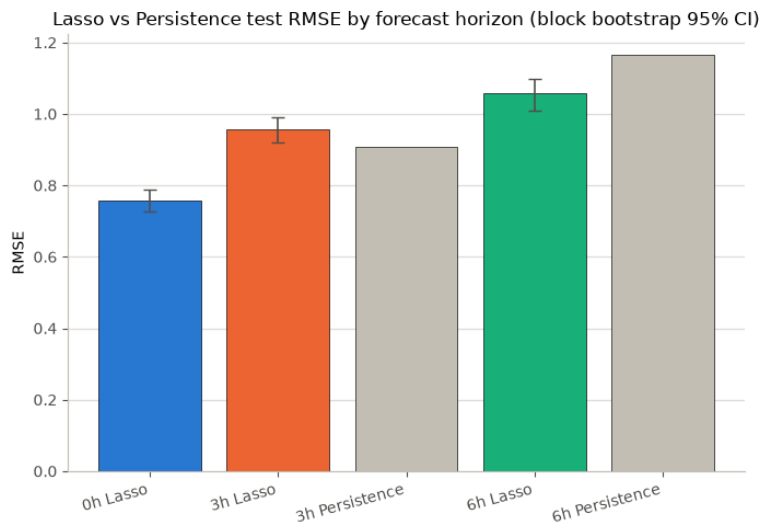


Figure 2 - RMSE for Lasso Regression and Kp Persistence (Multiple Time Horizons)

5. User Interface and Application

5.1 User Inputs and Features

The user-facing application developed in Streamlit provides an intuitive interface for interacting with space weather forecasts and visualizing model predictions. Users can select a nowcast or forecast horizon, and choose a city from a dropdown menu to get a visibility probability.

There is also an interactive map that allows users to manually adjust Kp values using a slider to simulate custom space weather scenarios. The visualization dynamically renders predicted auroral coverage based on the inputs. The underlying mathematical model responds dynamically to changes in solar orientation and geomagnetic storm intensity.

Figure 3 illustrates the effect of shifting subsolar longitude across 0.0 degrees, 90.0 degrees, and 180.0 degrees while holding geomagnetic activity constant at Kp = 2. The visualization models this physical behavior by compressing the sunward arc into a thin, poleward-shifted band, while expanding the nightside region into a broader arc displaced roughly 5 degrees equatorward. As the solar longitude rotates across the three panels, the compressed dayside section smoothly rotates around the geographic pole to remain aligned with the Sun, keeping the thickest, highest-probability viewing zone facing local midnight.

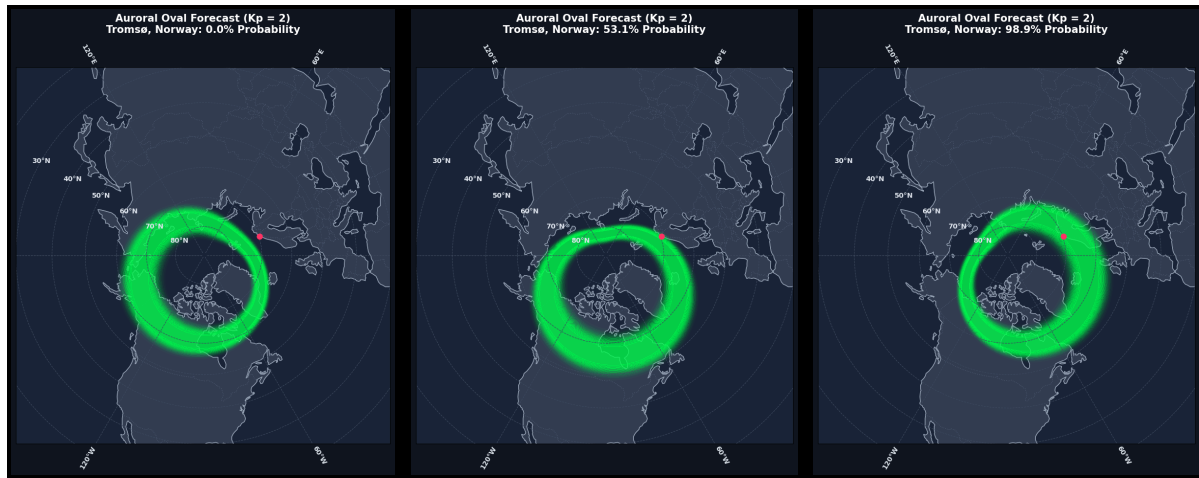


Figure 3 - Auroral Oval Asymmetry Across Solar Longitudes: Polar projections at Kp = 2 displaying nightside oval displacement and dayside arc compression across solar longitudes of 0.0°, 90.0°, and 180.0°.

Figure 4 displays the spatial impact of escalating geomagnetic storms across Kp values of 2, 5, and 8 at a fixed solar position (solar longitude = 180.0). During quiet conditions (Kp = 2), the auroral oval remains a narrow ring confined above 65 degrees North latitude, yielding low viewing probability for most cities. As storm activity intensifies to moderate levels (Kp = 5), the visualization expands the peak latitude equatorward and broadens the width of the band, pushing higher percent probability over Scandinavia and northern Canada. At severe storm levels (Kp = 8), the outer boundary of the oval expands even further toward mid-latitude regions, thickening the high-probability green band southward while widening the dark polar cap core.

As the Kp index increases, the area covered by the auroral oval increases, with the outer boundary of the oval moving to latitudes closer to the equator.

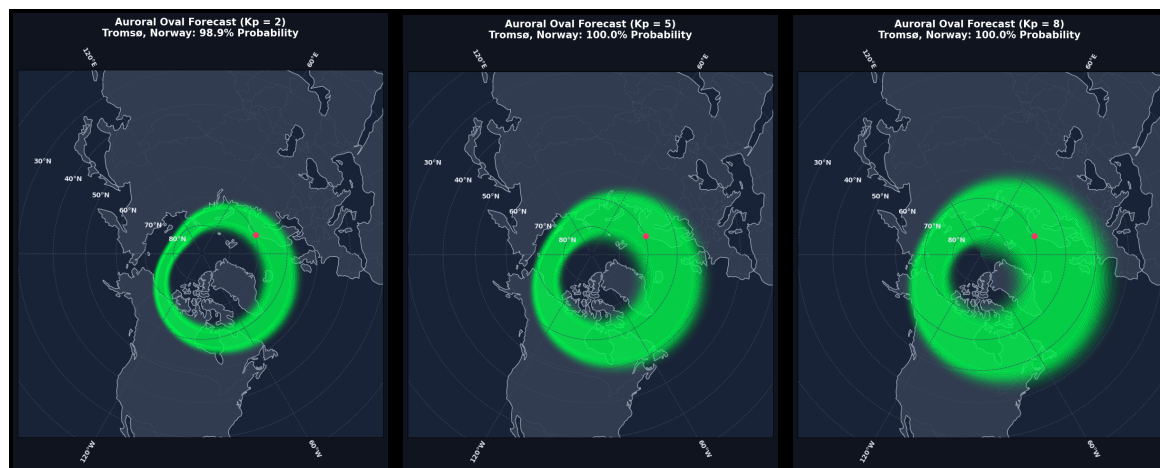


Figure 4 - Auroral Oval Expansion Across Geomagnetic Intensity Levels: Spatial forecast maps showing equatorward migration and coverage expansion across quiet (Kp = 2), moderate (Kp = 5), and severe (Kp = 8) storm conditions.

5.2 Real-Time Data Pipeline

Our final pipeline integrates real-time data with our trained Kp index model and the Streamlit application's aurora visualization:

1. **Retrieve real-time data:** Through the NOAA's JSON endpoints, 1-minute resolution geomagnetic data is pulled from satellites. These were then used to create a DataFrame with a form/structure similar to how it is with OMNIWeb. This is called when the user first accesses the Streamlit app and every 10 minutes after that, both to avoid excessive calling of the pipeline and considering the Kp index is calculated every 3 hours.
2. **Data Preprocessing:** We took the same steps described before to preprocess our data. This includes the 1-hour interpolation to address NaNs, as well as the log-transformations of a few columns to address the skewed distribution of respective features.
3. **Feature Engineering:** The preprocessed rows were then aggregated into the same 3-hour feature structure used during model training. For each prediction interval, we calculated the average, minimum, and maximum of each feature over the relevant hourly look-back periods. These features were then organized into the structure expected by our model.
4. **Model Prediction:** The data is now in a state ready to be fed into one of the trained Lasso regression models. The predicted Kp index for the selected time horizon is then displayed in the app along with other metrics such as forecast time and solar longitude, as well as the generated visualization of the auroral oval.

6. Discussion

In our project, we found that combining machine learning techniques, geomagnetic activity modeling, spherical trigonometry, and Gaussian location-based probability models can be an effective tool for generating nowcasts and short-term forecasts of geomagnetic storms and aurora visibility. For nowcasts, we found that LSTM neural networks trained with 24 hourly look-backs had the lowest prediction error with an MAE of 0.53. We found that Lasso regression with 4 hourly look-backs could produce similar results (MAE: 0.58) to LSTM and function as an effective prediction tool in a real-time data pipeline, running in 0.6 seconds end-to-end. Based on the MAE, our Lasso model differed from the actual Kp values by approximately 0.58 Kp units on average, which is only 6.4% of the Kp scale range. Training the Lasso model for 6-hour forecasts showed superior predictive power relative to the baseline of the Kp persistence model (MAE: 0.80 and 0.88, respectively). We were surprised to see the 3-hour forecast lagging slightly behind the persistence model, with a 0.05 difference in MAE. This suggests that the Kp persistence model is difficult to overtake for short horizons, but for forecasts farther into the future, its performance deteriorates more rapidly than our models.

Based on the values of the model coefficients shown in **Appendix A**, the top three most influential features in the Lasso regression model were the maximum solar wind speed (0-1 hour look-back), the maximum magnitude of the vector-averaged IMF (0-1 hour look-back), and the minimum proton density (0-1 hour look-back).

Our most significant challenges were data leakage risks and compute power constraints. Working with time series data requires careful handling to ensure that models do not learn patterns from future data. To mitigate this, we split our training, validation, and test sets chronologically before any imputation or feature engineering occurred. We chose the most recent 3 years for our test set, the next 3.5 years for the validation set, and the remaining years for the training set. We performed our parameter searches against the held-out validation set to ensure that leakage could not occur across folds with GridSearchCV. For the LSTM neural network, we did not possess GPU hardware to efficiently run grid searches and tuning, so we utilized a T4 GPU on Google Colab.

There is certainly room to improve our models, as results from previous studies on this topic have shown. With more time and resources, we would extend our LSTM model and train it for 3-hour and 6-hour forecasts. We would also focus our attention on refining the linear models for 3-hour forecasts, either by adding look-backs of the Kp values as features or by training an autoregressive model for comparison. Finally, we would explore a reduction in the number of features based on the coefficient values of the existing Lasso model or permutation importance for the LSTM model to see if this improves generalization. In terms of the Streamlit application, we would pursue integrating real-time cloud coverage alongside geomagnetic sight probability to deliver a combined ground visibility score for auroras.

7. Ethical Considerations

The main ethical concern with our project is ensuring that the users understand the limitations and reliability of our predictions. Although our preprocessing steps address sources of uncertainty in both historical and real-time datasets such as missing values, our Kp predictions nevertheless remain estimates rather than a source of truth. This is important considering that our RMSE increases as our forecast horizon also increases, meaning that longer-term predictions should be treated with greater caution. Inaccurate predictions could also cause users to incorrectly or unnecessarily travel in anticipation of an aurora prediction. Therefore, users should be made aware that the predictions and auroral-oval visualizations are merely estimates for informational purposes. We've added this as a disclaimer in our Streamlit app.

8. Statement of Work

- **Sophia:** Feature definitions, visualizations, mapping development, recording/editing video presentation, Streamlit UI design, Streamlit integrations, report writing
- **Tony:** Overall project coordination, feature engineering, model development and evaluation, report writing, report integration and editing
- **Kyle:** Data extraction and cleaning, data exploration, GitHub repo creation/organization, real-time data pipeline, Deepnote integrations, Streamlit integrations, report writing

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Appendix A - Feature List (Lasso Regression Models)

Ranked by absolute value of the coefficients

feature	coefficient
flow_speed_km_s_log_max_0_1h	0.600074
mag_avg_nt_log_max_0_1h	0.467124
proton_density_n_cc_log_min_0_1h	0.272073
proton_density_n_cc_log_max_0_1h	0.243627
proton_density_n_cc_log_avg_1_2h	-0.09401
flow_speed_km_s_log_max_3_4h	0.091986
bz_gsm_nt_min_0_1h	-0.07246
hour_sin	-0.067
mag_avg_nt_log_min_0_1h	0.065993
proton_density_n_cc_log_min_1_2h	-0.06501
bz_gsm_nt_max_0_1h	-0.05866
hour_cos	-0.05512
mag_avg_nt_log_avg_1_2h	-0.05152
bx_gsm_nt_max_0_1h	-0.04925
mag_avg_nt_log_avg_2_3h	-0.04641
bz_gsm_nt_avg_0_1h	-0.04635
day_cos	-0.04516
bz_gsm_nt_avg_3_4h	-0.03867
bx_gsm_nt_min_0_1h	0.036636
mag_avg_nt_log_max_1_2h	-0.03588
by_gsm_nt_avg_0_1h	0.03526
mag_avg_nt_log_min_3_4h	0.02932
proton_density_n_cc_log_max_2_3h	0.028987
proton_density_n_cc_log_min_3_4h	-0.02569
by_gsm_nt_min_0_1h	-0.02351

mag_avg_nt_log_min_2_3h	0.020976
by_gsm_nt_min_1_2h	-0.02034
bz_gsm_nt_avg_2_3h	-0.02012
bz_gsm_nt_min_2_3h	-0.01983
mag_avg_nt_log_min_1_2h	0.019314
bz_gsm_nt_max_3_4h	-0.0193
bx_gsm_nt_avg_1_2h	-0.01891
by_gsm_nt_min_3_4h	-0.01715
bx_gsm_nt_avg_3_4h	-0.01432
day_sin	-0.01385
bx_gsm_nt_min_1_2h	0.012838
mag_avg_nt_log_avg_0_1h	0.012768
mag_avg_nt_log_max_3_4h	-0.01262
by_gsm_nt_max_3_4h	0.010789
by_gsm_nt_max_1_2h	0.010127
bx_gsm_nt_min_2_3h	0.008721
flow_speed_km_s_log_min_1_2h	-0.00825
mag_avg_nt_log_max_2_3h	-0.00674
by_gsm_nt_avg_2_3h	0.006553
bz_gsm_nt_avg_1_2h	-0.00636
by_gsm_nt_avg_1_2h	0.005128
bx_gsm_nt_avg_0_1h	0.003079
bx_gsm_nt_max_1_2h	-0.00085
by_gsm_nt_avg_3_4h	0.000474
proton_density_n_cc_log_max_1_2h	0
bx_gsm_nt_avg_2_3h	0
by_gsm_nt_max_0_1h	0

proton_density_n_cc_log_avg_0_1h	0
flow_speed_km_s_log_min_0_1h	0
flow_speed_km_s_log_max_1_2h	0
flow_speed_km_s_log_avg_1_2h	0
bz_gsm_nt_max_1_2h	0
bz_gsm_nt_min_1_2h	0
flow_speed_km_s_log_avg_0_1h	0
by_gsm_nt_min_2_3h	0
mag_avg_nt_log_avg_3_4h	0
proton_density_n_cc_log_min_2_3h	0
flow_speed_km_s_log_max_2_3h	0
proton_density_n_cc_log_avg_2_3h	0
flow_speed_km_s_log_min_2_3h	0
flow_speed_km_s_log_avg_2_3h	0
by_gsm_nt_max_2_3h	0
bz_gsm_nt_max_2_3h	0
bx_gsm_nt_max_2_3h	0
bx_gsm_nt_max_3_4h	0
bx_gsm_nt_min_3_4h	0
bz_gsm_nt_min_3_4h	0
flow_speed_km_s_log_min_3_4h	0
flow_speed_km_s_log_avg_3_4h	0
proton_density_n_cc_log_avg_3_4h	0
proton_density_n_cc_log_max_3_4h	0